

DATA ANALYSIS AND VISUALIZATION

PROJECT REPORT

(Project Semester January–April 2026)

IPL PERFORMANCE ANALYSIS DASHBOARD

Submitted by

Alisha Chhabra

Registration No: 12413148

Bachelor of Technology (B.Tech)

Section: 3MO33

Course Code: INT 387

Under the Guidance of

Dr. Madhu Bala (31770)

Discipline of CSE/IT

Lovely School of Computer Science and Engineering

Lovely Professional University, Phagwara

April 2026

DECLARATION

I, ***Alisha Chhabra***, student of Bachelor of Technology (B.Tech) under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: April 2026

Registration No.: 12413148

Signature :


A handwritten signature in black ink, appearing to read 'Alisha Chhabra', is enclosed within a light gray rounded rectangular border.

Name of the student: Alisha Chhabra

CERTIFICATE

This is to certify that *Alisha Chhabra* bearing Registration no. *12413148* has completed *INT 387* project titled, “**IPL PERFORMANCE ANALYSIS DASHBOARD**” under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab.

Date: April 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my faculty mentor for their continuous support, guidance, and encouragement throughout the completion of this project. Their valuable suggestions helped me understand the concepts of data science and machine learning in a practical manner.

I would also like to thank Lovely Professional University for providing a platform to explore real-world datasets and apply theoretical knowledge to practical scenarios. This project has enhanced my understanding of data analysis, visualization, and predictive modeling.

Contents

INTRODUCTION.....	6
OBJECTIVES	8
DATASET DESCRIPTION.....	9
DATASET PREPROCESSING	11
DASHBOARD OVERVIEW.....	13
TEAM ANALYSIS	15
PLAYER ANALYSIS	17
VENUE ANALYSIS	19
PERFORMANCE METRICS	20
CHALLENGES FACED	21
RESULTS AND DISCUSSION.....	23
CONCLUSION	25
FUTURE SCOPE.....	26
REFERENCES.....	27
LINKEDIN	27

INTRODUCTION

In the contemporary world, data has become one of the most valuable resources across industries. The ability to collect, process, and analyze data plays a crucial role in decision-making and strategic planning. Data analytics enables organizations to uncover hidden patterns, identify trends, and derive meaningful insights from raw data.

The field of sports analytics has witnessed rapid growth, particularly in cricket, where performance analysis has become an integral part of team strategy and management. The Indian Premier League (IPL), being one of the most popular and competitive cricket leagues globally, generates vast amounts of data every season. This data includes detailed information about each ball bowled, runs scored, wickets taken, player contributions, and match outcomes.

The IPL dataset used in this project is highly granular, as it is recorded at the ball-by-ball level. While such detailed data provides an opportunity for deep analysis, it also presents challenges in terms of data handling, preprocessing, and aggregation. Without proper techniques, it is easy to derive incorrect or misleading insights from such data.

This project aims to address these challenges by developing an interactive dashboard using Microsoft Power BI. The dashboard is designed to transform raw IPL data into a structured and visually appealing format that allows users to explore insights related to team performance, player contributions, scoring patterns, and venue-based characteristics.

The key focus of this project is not only on visualization but also on ensuring logical correctness in data analysis. Special attention has been given to handling data granularity, applying appropriate aggregation techniques, and designing meaningful metrics using DAX (Data Analysis Expressions).

Through this project, various analytical perspectives have been explored, including:

- Team-wise performance trends across matches and seasons
- Player impact using Player of the Match awards
- Run progression across overs to understand match flow
- Boundary distribution across venues to analyse pitch behaviour
- Performance comparison using strike rate and economy rate

The dashboard developed in this project provides an intuitive interface that enables users to interact with the data, apply filters, and derive insights dynamically. It demonstrates how complex datasets can be simplified through effective visualization and analysis techniques.

Overall, this project highlights the importance of data preprocessing, analytical thinking, and visualization design in building effective data-driven solutions.

OBJECTIVES

The main objectives of this project are:

1. To analyze IPL ball-by-ball data and extract meaningful insights.
2. To understand team performance trends across different seasons.
3. To evaluate player contributions using Player of the Match awards.
4. To analyze scoring patterns using run progression across overs.
5. To study venue-based performance using boundary distribution.
6. To compare batting and bowling efficiency using strike rate and economy rate.
7. To develop an interactive dashboard using Power BI.
8. To apply DAX functions for accurate data aggregation and calculations.
9. To present complex data in a simple and understandable visual format.

DATASET DESCRIPTION

The dataset used in this project consists of IPL ball-by-ball data collected from publicly available cricket data sources. The dataset covers multiple IPL seasons from 2008 to 2023.

The dataset is structured and contains detailed information about each delivery bowled in a match. Each row in the dataset represents a single ball, making it highly granular.

Key attributes in the dataset include:

- Match ID
- Batting Team
- Bowling Team
- Player of the Match
- Runs scored (batsman runs, extras)
- Wickets
- Overs and ball numbers
- Venue and city

Due to the ball-by-ball nature of the dataset, a single match appears multiple times, which requires careful handling during analysis to avoid duplication errors.

The dataset is large and complex, making it suitable for advanced data analysis and visualization tasks.

Match ID	Date	Innings	Batting Team	Bowling Team	over	Ball Number	Valid Ball	Extras	Total Runs	Bowler Runs
598011	13 April 2013	1	Mumbai Indians	Pune Warriors	0	0.1	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	8	8.2	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	8	8.4	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	9	9.2	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	9	9.3	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	9	9.4	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	10	10.2	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	10	10.3	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	10	10.5	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	10	10.6	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	11	11.3	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	11	11.5	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	11	11.6	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	10	10.1	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	7	7.5	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	2	2.6	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	3	3.2	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	3	3.4	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	3	3.6	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	5	5.3	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	5	5.6	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	6	6.1	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	6	6.2	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	6	6.4	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	12	12.3	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	17	17.5	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	18	18.2	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	18	18.4	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	17	17.3	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	13	13.1	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	13	13.4	1	0	1	1
598037	29 April 2013	1	Mumbai Indians	Kings XI Punjab	13	13.6	1	0	1	1

Sample View of IPL Ball-by-Ball Dataset

DATASET PREPROCESSING

Data preprocessing is a crucial step in the data analysis process, especially when working with highly granular datasets such as ball-by-ball cricket data.

The preprocessing of the dataset was performed using Power Query in Power BI. The following steps were carried out:

1. Data Cleaning:

- Removed null values and incomplete records.
- Standardized column formats and corrected inconsistencies.

2. Handling Granular Data:

- Since each match is represented by multiple rows, aggregation techniques were applied carefully.
- Used DISTINCTCOUNT instead of COUNT to avoid duplication in match-based analysis.

3. Data Transformation:

- Converted categorical data into usable formats.
- Created calculated columns for better analysis.

4. Feature Engineering:

- Developed measures for total runs, wickets, and averages.
- Calculated strike rate and economy rate using DAX formulas.

5. Data Validation:

- Verified results to ensure accuracy.
- Cross-checked aggregated values to avoid misleading insights.

Proper preprocessing ensured that the dataset was clean, structured, and ready for analysis.

Queries [1]

×

✓

fx

Table.TransformColumnTypes(*Added Custom4,{"Is Wicket", Int64.Type})

IPL

	Player Dismissed	fielders	Target Runs	Player of the Match	Winner	Toss Winner	Toss
1	None		null	null Akash Madhwal	Rajasthan Royals	Rajasthan Royals	field
2	None		null	null MEK Hussey	Chennai Super Kings	Chennai Super Kings	bat
3	None		null	null Sohail Tanvir	Rajasthan Royals	Chennai Super Kings	bat
4	None		null	null Akash Madhwal	Rajasthan Royals	Rajasthan Royals	field
5	None		null	null MEK Hussey	Chennai Super Kings	Chennai Super Kings	bat
6	None		null	null MS Dhoni	Chennai Super Kings	Chennai Super Kings	bat
7	None		null	null ML Hayden	Chennai Super Kings	Mumbai Indians	field
8	None		null	null Sohail Tanvir	Rajasthan Royals	Chennai Super Kings	bat
9	None		null	null ML Hayden	Chennai Super Kings	Mumbai Indians	field
10	None		null	null AC Gilchrist	Deccan Chargers	Deccan Chargers	field
11	S Dube	YBK Jaiswal		null AC Gilchrist	Rajasthan Royals	Rajasthan Royals	field
12	None		null	null MEK Hussey	Chennai Super Kings	Chennai Super Kings	bat
13	None		null	148 JDP Oram	Chennai Super Kings	Kolkata Knight Riders	bat
14	None		null	null MEK Hussey	Chennai Super Kings	Chennai Super Kings	bat
15	None		null	null MS Dhoni	Chennai Super Kings	Chennai Super Kings	bat
16	None		null	null MEK Hussey	Chennai Super Kings	Chennai Super Kings	bat
17	None		null	148 JDP Oram	Chennai Super Kings	Kolkata Knight Riders	bat
18	None		null	null ML Hayden	Chennai Super Kings	Mumbai Indians	field
19	None		null	null Sohail Tanvir	Rajasthan Royals	Chennai Super Kings	bat
20	None		null	null ML Hayden	Chennai Super Kings	Mumbai Indians	field
21	None		null	null ML Hayden	Chennai Super Kings	Mumbai Indians	field
22	None		null	null AC Gilchrist	Deccan Chargers	Deccan Chargers	field
23	None		null	null AC Gilchrist	Deccan Chargers	Deccan Chargers	field
24	None		null	null MEK Hussey	Chennai Super Kings	Chennai Super Kings	bat
25	None		null	188 MS Dhoni	Chennai Super Kings	Chennai Super Kings	field
26	None		null	null Akash Madhwal	Rajasthan Royals	Rajasthan Royals	field
27	None		null	null MEK Hussey	Chennai Super Kings	Chennai Super Kings	bat
28	None		null	null MS Dhoni	Chennai Super Kings	Chennai Super Kings	bat

28 COLUMNS, 999+ ROWS

Column profiling based on top 1000 rows

Query Settings

PROPERTIES

Name

IPL

All Properties

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Removed Columns

Changed Type1

Renamed Columns

Changed Type2

Replaced Value

Replaced Value1

Changed Type3

Replaced Value2

Replaced Value3

Replaced Value4

Added Custom

Added Custom1

Removed Columns1

Added Custom2

Removed Columns2

Added Custom3

Changed Type4

Filtered Rows

Sorted Rows

Renamed Columns1

Data Cleaning and Transformation using Power Query

DASHBOARD OVERVIEW

The IPL Performance Analysis Dashboard developed in this project is a multi-page interactive dashboard designed using Microsoft Power BI. The primary objective of the dashboard is to present complex cricket data in a structured, intuitive, and visually engaging manner.

The dashboard is divided into multiple sections, each focusing on a specific aspect of analysis. This modular approach allows users to navigate through different perspectives of the dataset without confusion.

The main pages of the dashboard include:

1. Overview Dashboard:

This page provides a summary of key performance indicators (KPIs), such as total runs, total wickets, and overall match statistics. It gives users a quick understanding of the dataset at a glance.

2. Team Analysis:

This section focuses on team performance using visualizations such as area charts and decomposition trees. It helps in understanding how teams perform across different phases of the match.

3. Player Performance:

This page highlights individual player contributions using Player of the Match awards and other performance metrics.

4. Venue-Based Analysis:

This section uses map visualization to analyze boundary distribution across different cities, providing insights into pitch behavior and scoring patterns.

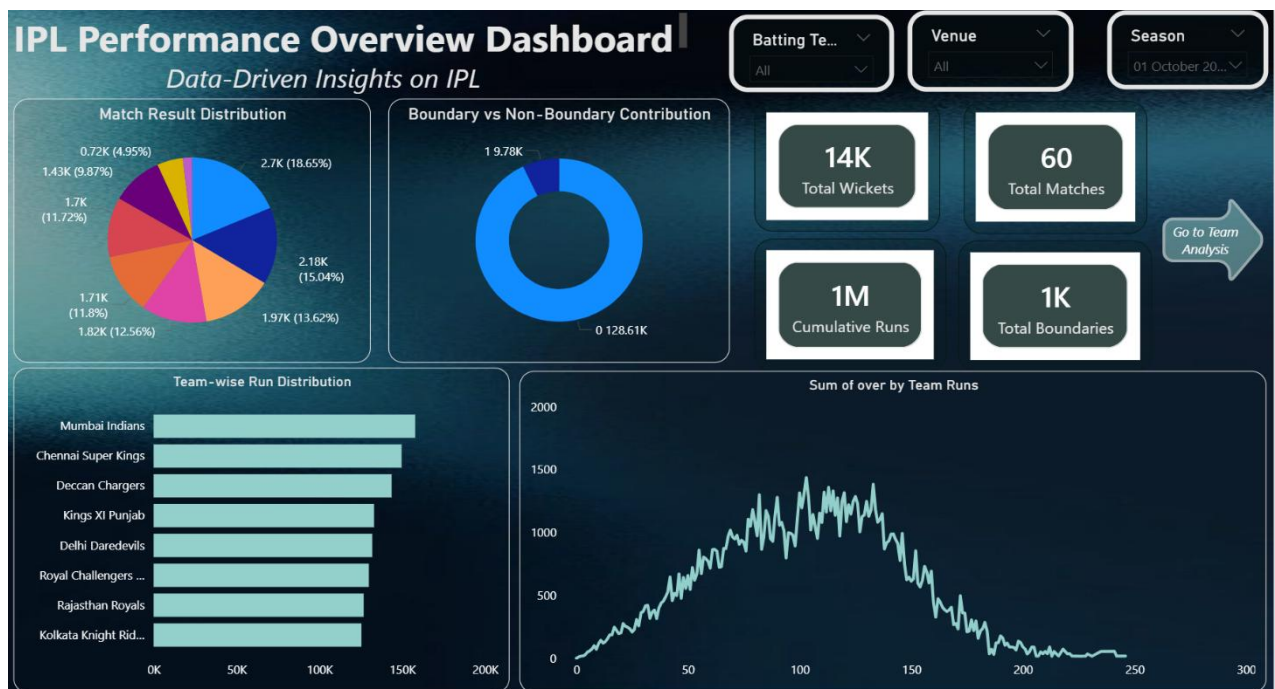
The dashboard follows a consistent dark-themed design to enhance visual clarity and reduce eye strain. Careful attention has been given to layout, alignment, and spacing to ensure that the dashboard remains clean and uncluttered.

Interactive features such as slicers and navigation buttons have been incorporated to allow users to filter data dynamically and switch between different views. These features improve user experience and make the dashboard more engaging.

One of the key aspects of the dashboard design is the selection of appropriate visualizations. Each chart has been chosen based on the type of data and the insight it provides. For example:

- Area charts are used for run progression analysis
- Scatter plots are used for comparing strike rate and economy rate
- Maps are used for geographical analysis
- Tables are used for detailed player information

Overall, the dashboard effectively combines data preprocessing, analytical modeling, and visualization techniques to provide meaningful insights into IPL performance.



Overview Dashboard of IPL Analysis

TEAM ANALYSIS

Team performance analysis plays a crucial role in understanding match outcomes and identifying strengths and weaknesses of different teams.

In this project, team analysis was performed using multiple visualizations, including area charts and decomposition trees.

The area chart was used to analyse run progression across overs. This visualization helps in understanding how teams accumulate runs throughout the innings. It was observed that teams often accelerate scoring during middle overs, which significantly influences match momentum.

The decomposition tree was used to break down team performance into multiple contributing factors. This advanced visualization allows users to explore data hierarchically and identify patterns that may not be visible in traditional charts.

The analysis revealed that team performance varies based on match conditions, strategy, and player contributions. Some teams show consistent performance across matches, while others exhibit variability depending on specific conditions.

Overall, team analysis provides valuable insights into scoring patterns, performance trends, and match dynamics.



Run Progression Across Overs



Advanced Team Analysis using Decomposition Tree

PLAYER ANALYSIS

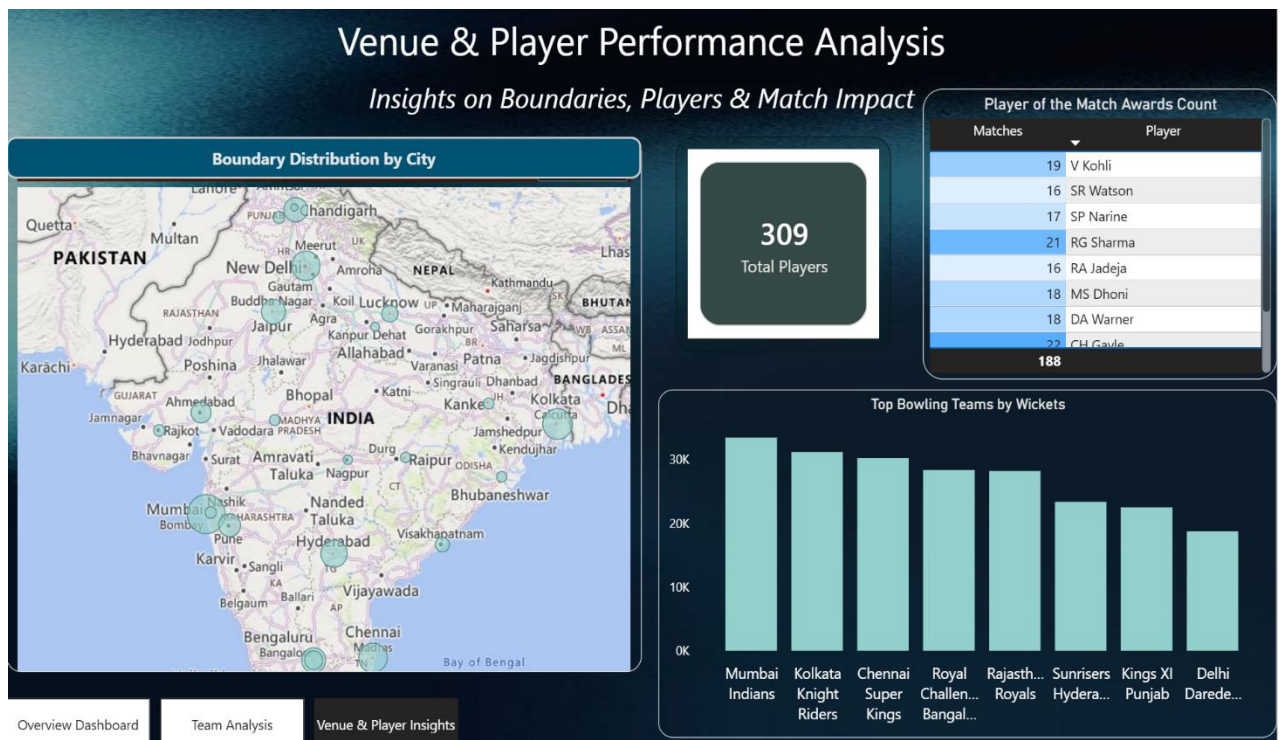
Player performance is one of the most important aspects of cricket analytics, as individual contributions often determine match outcomes.

In this project, player performance was analyzed using the Player of the Match metric, which reflects match-winning contributions.

Since the dataset is recorded at the ball-by-ball level, each match appears multiple times. To accurately calculate the number of awards received by each player, distinct count aggregation was used.

The analysis identifies players who have consistently performed well and contributed significantly to their teams' success. These players can be considered key performers or match winners.

This analysis provides a deeper understanding of player impact beyond basic statistics.



Player of the Match Awards Count	
Matches	Player
19	V Kohli
16	SR Watson
17	SP Narine
21	RG Sharma
16	RA Jadeja
18	MS Dhoni
18	DA Warner
22	CH Gayle
188	

Top Players by Match Awards

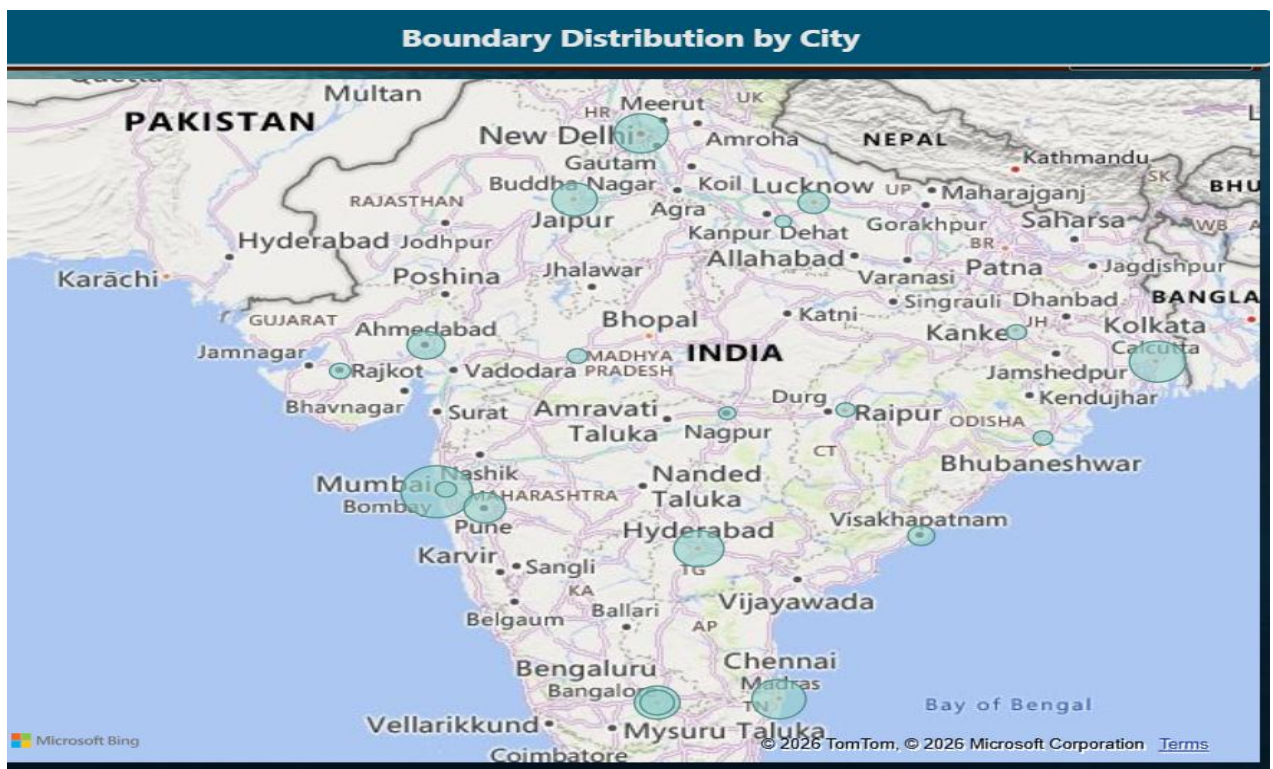
VENUE ANALYSIS

Venue-based analysis helps in understanding how different playing conditions affect match outcomes.

In this project, map visualization was used to analyse boundary distribution across cities. The size of the bubbles on the map represents the frequency of boundaries scored at each venue.

The analysis shows that certain venues consistently produce high-scoring matches, indicating favourable batting conditions. These insights can be useful for teams to plan strategies based on venue characteristics.

Venue analysis highlights the importance of external factors such as pitch conditions and ground dimensions in influencing match performance.



Boundary Distribution Across Cities

PERFORMANCE METRICS

Performance metrics such as strike rate and economy rate were analyzed to evaluate batting and bowling efficiency.

Strike rate measures how quickly a batsman scores runs, while economy rate measures how efficiently a bowler controls runs.

A scatter plot was used to compare these metrics, providing a visual representation of performance balance.

This analysis helps in identifying players who perform efficiently under different conditions and contributes to a deeper understanding of match dynamics.



Strike Rate vs Economy Analysis

CHALLENGES FACED

The development of the IPL Performance Analysis Dashboard involved several challenges due to the complexity and nature of the dataset. Addressing these challenges was essential to ensure accurate analysis and meaningful insights.

1. Handling Highly Granular Data:

The IPL dataset used in this project is recorded at a ball-by-ball level, meaning that each match consists of multiple rows. This level of granularity significantly increases the size of the dataset and introduces complexity in data handling. A single match appearing multiple times made it difficult to perform match-level analysis without proper aggregation.

2. Data Duplication and Aggregation Issues:

Since each match is represented by numerous rows, direct aggregation functions such as COUNT often resulted in incorrect values. For example, counting match IDs without considering duplication led to inflated results. To overcome this, DISTINCTCOUNT was used to ensure accurate representation of match-level metrics.

3. Data Cleaning and Consistency:

The dataset required extensive cleaning to remove inconsistencies, null values, and incorrect data entries. Ensuring uniformity across columns such as team names, player names, and venues was necessary to maintain data integrity.

4. Selection of Appropriate Visualizations:

Choosing the right visualization for each type of analysis was a challenging task. Incorrect visual choices could lead to misinterpretation of data. Careful consideration was required to match each visualization with the nature of the data and the intended insight.

5. Maintaining Dashboard Simplicity:

While the dataset allowed for multiple analyses, including too many visuals could lead to clutter and reduce readability. A balance had to be maintained between providing detailed insights and keeping the dashboard clean and user-friendly.

6. Performance Optimization:

Working with a large dataset sometimes affected the performance of the dashboard. Efficient data modeling and optimized DAX calculations were necessary to ensure smooth interaction.

7. Ensuring Logical Accuracy:

One of the most critical challenges was ensuring that all calculations and visualizations were logically correct. Even a small mistake in aggregation or filtering could lead to misleading insights, making validation an essential part of the process.

Overall, overcoming these challenges helped in building a robust, accurate, and user-friendly dashboard.

RESULTS AND DISCUSSION

The analysis performed in this project provided several meaningful insights into IPL matches, teams, players, and venues. The results demonstrate the effectiveness of data visualization in transforming complex datasets into understandable information.

1. Team Performance Insights:

The team analysis revealed that scoring patterns vary significantly across different phases of the match. The run progression analysis indicated that middle overs play a crucial role in determining match outcomes. Teams that perform well during this phase often gain a competitive advantage.

2. Player Contribution Analysis:

The Player of the Match analysis highlighted key players who consistently contribute to match outcomes. These players demonstrate strong performance under pressure and play a vital role in their team's success.

3. Venue-Based Observations:

The map visualization showed that certain venues have a higher frequency of boundaries, indicating favorable batting conditions. This suggests that pitch characteristics and ground dimensions significantly influence match performance.

4. Performance Metrics Evaluation:

The comparison of strike rate and economy rate provided insights into batting and bowling efficiency. Players with high strike rates contribute to faster scoring, while bowlers with low economy rates effectively restrict runs.

5. Impact of Data Granularity:

The project emphasized the importance of handling granular data correctly. Proper aggregation techniques were essential to ensure accurate results, highlighting the significance of data preprocessing in analytics.

6. Visualization Effectiveness:

The use of different types of visualizations enhanced the interpretability of data. Interactive elements allowed users to explore insights dynamically, improving user experience.

The results of this project demonstrate how data analytics can be applied to sports data to derive meaningful conclusions and support decision-making.

CONCLUSION

The IPL Performance Analysis Dashboard successfully demonstrates the application of data analytics and visualization techniques in transforming complex datasets into meaningful insights.

The project highlights the importance of data preprocessing, correct aggregation, and effective visualization design in building reliable analytical solutions. By working with highly granular ball-by-ball data, the project provided practical experience in handling large datasets and ensuring logical accuracy.

The dashboard developed in this project provides a comprehensive view of IPL performance, including team trends, player contributions, scoring patterns, and venue-based analysis. It enables users to explore data interactively and gain insights that are not easily visible in raw datasets.

Through this project, a deeper understanding of tools such as Power BI, Power Query, and DAX was achieved. It also emphasized the importance of analytical thinking and problem-solving in data-driven projects.

Overall, the project successfully meets its objectives and demonstrates the potential of data analytics in sports and other domains.

FUTURE SCOPE

The IPL Performance Analysis Dashboard can be further enhanced and extended in several ways to increase its functionality and analytical capabilities.

1. Integration with Real-Time Data:

Future work can include integrating live IPL data to provide real-time analysis and insights during ongoing matches.

2. Predictive Analytics:

Machine learning models can be applied to predict match outcomes, player performance, and team strategies based on historical data.

3. Advanced Player Analytics:

Additional metrics such as player consistency, pressure performance, and contribution in different match situations can be analyzed.

4. Web-Based Deployment:

The dashboard can be deployed as a web application, making it accessible to a wider audience.

5. Enhanced Visualizations:

More advanced and customized visualizations can be added to improve user experience and provide deeper insights.

6. Integration with Other Data Sources:

Combining IPL data with external datasets such as weather conditions and player fitness can provide more comprehensive analysis.

These enhancements would make the dashboard more powerful and useful for analysts, teams, and cricket enthusiasts.

REFERENCES

- [1] IPL Dataset, Public Cricket Data Sources, Available Online.
- [2] Microsoft, “Power BI Documentation,” Available: <https://learn.microsoft.com>
- [3] Microsoft, “DAX Functions Reference,” Available: <https://learn.microsoft.com>
- [4] Data Analytics Tutorials, Various Online Learning Platforms.
- [5] Sports Analytics Research Articles and Publications.

LINKEDIN

URL to LinkedIn